

a tensor: two slices

$$\frac{\partial f_1}{\partial A} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial f_2}{\partial A} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \end{bmatrix}$$

shape $2 \times 2 \times 3$

flatten A $\rightarrow$

one Jacobian: 2×6

$A_{11} \ A_{12} \ A_{13} \ A_{21} \ A_{22} \ A_{23}$

$$\begin{bmatrix} 1 & 2 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 3 \end{bmatrix}$$

row i is non-zero only on row i of A , and there it equals $\mathbf{x} = [1, 2, 3]$